O–B–491b
28 MAY 1954
SUPERSEDING
Int. Fed. Spec. O–B–491a (Navy–Ships)
22 April 1952 and
Fed. Spec. O–B–491
6 February 1934

FEDERAL SPECIFICATION

BLUING, LAUNDRY

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 **Scope.**—This specification covers laundry bluing (nonsour, sour) for general laundry use to neutralize the slightly yellow shade of fabrics by coloring them with a faint blue tint.

1.2 **Classification.**

1.2.1 *Types and classes.*—Laundry bluing shall be furnished in the following types and classes, as specified (see 6.2):

Type I.—Nonsour
Class A.—Balls, blocks, or cubes
Class B.—Powder or tablet

Type II.—Sour, powder or tablet

Type III.—Liquid (nonsour or sour)

2. APPLICABLE SPECIFICATIONS, STANDARD, AND OTHER PUBLICATIONS

2.1 The following specifications and standard, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

P–S–566—Soap; Chip.
CCC–F–451—Flannel; Canton.
CCC–S–271—Sheeting; Cotton, Bleached, Wide.
CCC–T–191—Textile Test Methods.

Military Standard:

MIL–STD–105—Sampling Procedures and Tables for Inspection by Attributes.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under "General Information" in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the GSA Regional offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.

(Copies of Military specifications and standards, required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

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2.2 **Other publications.** — The following publications, of the issues in effect on date of invitation for bids, form a part of this specification:

Post Office Department:
Postal Regulations.

(Application for copies should be addressed to the Post Office Department, Washington 25, D. C.)

Consolidated Classification Committee:
Consolidated Freight Classification Rules.

(Application for copies should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago 6, Ill.)

American Trucking Association:
Motor Freight Classification Rules.

(Application for copies should be addressed to the American Trucking Association, 1424 16th St., NW., Washington 6, D. C.)

2.3 Specifications and publications applicable only to individual departments are listed in section 7.

3. REQUIREMENTS

3.1 Types I and II, nonsour and sour.

3.1.1 The bluing shall tint clothes to a bright white and shall be fugitive to washing.

3.1.2 *Tinctorial strength.*—When a standard sample is furnished by the procuring activity the tinctorial strength of deliveries shall be not less than that of the standard sample when tested as specified in 4.3.1.1.

3.2 Type I, nonsour.

3.2.1 *Class A, ball, blocks or cubes.*—The class A bluing shall be a good quality, insoluble, ultramarine blue.

3.2.1.1 *Color.*—The color of the bluing shall be reddish blue when determined as specified in 4.3.2.2.

3.2.1.2 *Fineness of suspended particles.*—The particles of bluing shall be of such size that when suspended in water they will require not less than 2 hours to completely settle when tested as specified in 4.3.2.3.

3.2.2 *Class B, powder or tablet.*—The class B bluing shall be water soluble aniline blues.

3.2.2.1 *Color.*—The color of the solution shall be reddish blue when determined as specified in 4.3.3.2.

3.3 Type II, sour, powder or tablet.—The type II bluing shall conform to 3.1, 3.2.2, and 3.2.2.1.

3.4 Type III, liquid (nonsour or sour).

3.4.1 *Material.* — Liquid laundry bluing shall consist of a solution of water-soluble aniline blue dye. It shall be of high quality, free from adulterants such as sodium chloride, sodium sulfate, and sugar, and shall be fugitive to washing. When diluted with water in the proportions recommended by the supplier, the bluing shall yield a stock solution which, when applied under the usual conditions of laundry operation, shall tint white goods to a bright white.

3.4.2 *Color.*—The color of the bluing shall be either red-blue or blue-blue when determined as specified in 4.3.5.2.

3.4.3 *pH.*—The pH of the bluing shall be not less than 6.5 and not greater than 8.5 when determined as specified in 4.3.5.3.

3.4.4 *Insoluble dye.* — There shall be no evidence of insoluble dye when tested as specified in 4.3.5.4.

3.4.5 *Freedom from pigment colors and methylene blue.* — Neither pigment colors

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such as ultramarine blue, prussian blue, nor methylene blue shall be present when determined as specified in 4.3.5.5.

3.4.6 *Resistance to decomposition by acid and alkali.*—There shall be no evidence of decomposition, such as curd, precipitate, or sediment of bluing allowed to remain in contact with acid and alkali when tested as specified in 4.3.5.6.

3.4.7 *Fastness to ironing.* — Blue cotton fabric shall not fade appreciably nor turn yellow on ironing when tested as specified in 4.3.5.7.

3.4.8 *Ease of removal in subsequent washing.*—The bluing shall be capable of being washed out completely from cotton fabric when tested as specified in 4.3.5.8. The blued, washed fabric shall be no more gray or yellow than a similarly treated specimen of unblued control cloth.

3.4.9 *Tinctorial strength.*—When tested as specified in 4.3.5.9, the bluing shall have Coleman Spectrophotometer readings not greater than 30 in acid solution, 45 in neutral solution, and 55 in alkaline solution. Minimum readings shall be 200 for the acid solution, 120 for the neutral solution, and 100 for the alkaline solution when the Klett-Summerson Photoelectric Colorimeter is used.

3.4.10 *Suitability in acid and alkaline solution.*—Based on the readings obtained in the test for tinctorial strength (see 4.3.5.9), the Coleman reading in alkaline solution shall show a decrease in color intensity of not more than 15 percent from the reading obtained in neutral solution. Using the Klett-Summerson Colorimeter, the maximum permitted decrease in alkaline solution compared to the neutral solution shall be 20 percent. In acid solution, using either instrument, there shall be no decrease in color intensity compared to that obtained in neutral solution. Percent decrease shall be calculated as follows:

$$\text{Percent decrease} = \frac{R_{alk} - R_n}{R_n} \times 100,$$

where:

R_{alk} is reading in alkaline solution.
R_n is reading in neutral solution.

3.5 **Workmanship.** — The bluing shall be completely dissolved and processed in accordance with good commercial practice for manufacture of this type of material. The product shall be stable in a sealed container and not subject to abnormal change with age.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 **Sampling.**—Each case or large container shall be carefully identified by manufacturer's batch (or control lot) number, until ultimate action is taken as to acceptance or rejection of material offered for delivery. Individual samples shall not be mixed, and shall be placed in separate airtight and watertight containers, which shall be nearly filled, covered and sealed to prevent atmospheric effects, and labelled completely with information on the lot and batch number, date of sampling, contract number, and applicable specification.

4.1.1 *Sampling lot.*—For purposes of sampling, a lot shall consist of all bluing of one type and class from one batch or "control lot". In cases in which, in the opinion of the Government inspector, it is not practicable to define the sampling lot in this manner, the lot shall consist of all bluing of one type and class offered for delivery at one time.

4.1.2 *Sampling for inspection of filled containers.*—A random sample of filled containers shall be taken from each lot by the Government inspector in accordance with Military Standard MIL-STD-105 at Inspec-

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tion Level I, and Acceptable Quality Level = 2.5 percent defective to verify compliance with this specification regarding fill, closure, marking and other requirements not involving tests.

4.1.3 *Sampling procedure for lot acceptance tests.*

4.1.3.1 *Types I and II.*—From each inspection lot, the Government inspector shall take two containers at random. From each of the two containers two one-ounce specimens shall be taken and placed in separate containers for the tests specified in 4.2.2.

4.1.3.2 *Type III.* — From each inspection lot, the Government inspector shall take at random two one-quart bottles of bluing for tests specified in 4.2.2.

4.1.4 *Sampling procedure for production check tests.*—In addition to the sample selected for lot acceptance tests, the Government inspector shall select a set of three additional samples, 3 ounces for type I or II or 3 quarts for type III, representative of the first lot offered for delivery under a contract or order, and therafter similar sets from any one lot in each group of 10 successive lots. Each sample shall be divided into three portions. One portion shall be forwarded to the Government laboratory designated by the procuring agency, one portion shall be delivered to the contractor, and one shall be held by the Government inspector for retests in case of dispute.

4.2 Inspection tests.

4.2.1 *Inspection of filled containers.*—Each sample filled container selected in accordance with 4.1.2 shall be examined by the Government inspector for defects of the container and the closure, for evidence of leakage, and for unsatisfactory markings; each sample filled container shall also be weighed to determine the amount of the contents. Any container in the sample having one or more defects, or under required fill, shall be rejected, and if the number of defective containers in any sample exceeds the acceptance number for the appropriate sampling plan of Military Standard MIL-STD-105, the lot represented by the sample shall be rejected. Rejected lots may be resubmitted for acceptance tests provided that the contractor has removed or repaired all nonconforming containers.

4.2.2 *Lot acceptance tests.*—The samples selected in accordance with 4.1.3 shall be subjected to the tests specified in 4.3.

4.2.2.1 *Place of test.* — Unless otherwise specified in the contract or order, lot acceptance tests shall be conducted at the contractor's plant, by the contractor in the presence and under the supervision of the Government inspector. Contractors shall provide the necessary laboratory facilities satisfactory to the inspector, or shall engage the services of a commercial testing laboratory acceptable to inspector.

4.2.2.2 *Action in case of failure.*—If any one of the samples tested is found to be not in conformance with this specification, the lot which it represents shall be rejected. A rejected lot may be resubmitted for Government inspection only after the manufacturer having been informed of the reasons for rejection, has so reworked the entire lot as to remove or correct all nonconforming material.

4.2.3 *Production check tests (at a Government laboratory).*—The sample selected in accordance with 4.1.4 shall be subjected to the appropriate tests specified in 4.3. If the type III bluing is not in conformance with 3.4.5, the sample need not be subjected to other tests before reporting the lot as unsatisfactory.

4.2.3.1 *Action in case of failure.*—Acceptance of the first lot offered for delivery under

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a contract or order shall be withheld until a satisfactory report is received on the production check test sample. Thenceforth, except as hereinafter specified, acceptance and rejection of lots shall normally be on the basis of the sampling and inspection specified in 4.1.3 and 4.2.2, and acceptance shall not be withheld pending receipt of test reports on production check test samples. However, upon receipt of an unsatisfactory test report on a production check test sample, the Government inspector shall select additional samples from every subsequent lot offered for delivery. The samples so selected shall be submitted to the Government laboratory, designated by the activity concerned, and shall there be subjected to test or tests wherein failure was observed. Lots shall then be accepted only upon receipt of satisfactory test report on the sample so selected. Additional testing shall be discontinued and lot acceptance returned to the normal basis when three successive lots have been accepted.

4.3 Test procedures.

4.3.1 *Types I and II.*

4.3.1.1 *Tinctorial strength.*—Bleached cotton sheeting shall be laundered with hot water and soap conforming to Federal Specification P–S–566. The sheeting shall be thoroughly rinsed with hot water and shall be dried. Specimens of the sheeting, 7 inches square, shall be cut for the test. (It is essential that these specimens be of the same size.) The same weights of the standard and of the sample under test (about 0.1 gm. of type I, class B, and type II, and about 1.0 gm. of type I, class A) shall be weighed out and transferred to 250-milliliter volumetric flasks. About 100 ml. of hot, distilled water shall be added and the flasks shall be whirled to dissolve or suspend the bluing. The solution shall be cooled to room temperature and shall be made up to the mark with distilled water. The flask shall be stoppered and shall be mixed thoroughly by shaking. The 7- by 7-inch specimens of sheeting shall be placed in 1 pint fruit jars, and 100 ml. of distilled water at room temperature shall be added. The cloths shall be wet thoroughly by shaking the jars. To one jar shall be added a 50-ml. aliquot of the *thoroughly mixed* solution or suspension of the standard sample, and to the other jar shall be added a 50-ml. aliquot of the *thoroughly mixed* solution or suspension of the sample under test. The jars shall be closed and shaken vigorously with same motion for 15 minutes. The blued specimens shall be removed from the jars and shall be dried under the same conditions. The sample under test has equal or greater tinting strength than the standard sample if the cloth blued with it is as dark or darker than that blued with the standard sample.

4.3.2 *Type I, class A.*

4.3.2.1 *Qualitative test.*—A portion of the sample shall be treated with dilute hydrochloric acid. If the material is ultramarine blue, the color will be destroyed with the evolution of hydrogen sulphide.

4.3.2.2 *Color.*—A portion of the sample shall be placed in a small bag made of two layers of canton flannel, conforming to Federal Specification CCC–F–451. The bag shall be securely tied and shall then be agitated through a volume of water until a faint blue-colored uniform suspension is obtained. The suspension shall be poured into a long test tube and the color shall be observed by looking down into the tube against a white background. The color observed shall be a reddish blue.

4.3.2.3 *Fineness of suspended particles.*—The intensity of color of the suspension used in the preceding test shall be increased by agitating the bag through the liquid until the blue color is very dense. The suspension shall be allowed to stand for 2 hours.

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4.3.3 *Type I, class B.*

4.3.3.1 *Solubility.*—One gm. of the bluing shall be dissolved in 100 ml. of distilled water at 90° C. (194° F.). Solution shall be rapid and complete, and there shall be no indication of settling out of solution after standing at room temperature for 24 hours.

4.3.3.2 *Color.* — The solution specified in 4.3.3.1 shall be diluted with distilled water until it is a pale blue color. The color of the solution shall be observed by looking down into a long test tube against a white background. The color observed shall be a reddish blue.

4.3.4 *Type II.*

4.3.4.1 *Solubility.*—One gm. of the bluing shall be dissolved in 100 ml. of distilled water at 90° C. (194° F.) and 1 ml. of glacial acetic acid shall be added. Solution shall be rapid and complete. There shall be no indication of sediment after allowing the solution to stand at room temperature for 24 hours.

4.3.4.2 *Color.* — The solution specified in 4.3.4.1 shall be diluted with distilled water until it is a pale-blue color. The color of the solution shall be observed as specified in 4.3.3.2.

4.3.5 *Type III.*

4.3.5.1 *Stock items.*

4.3.5.1.1 *Stock solution.*—One thousand ml. of stock solution shall be prepared by diluting the bluing with distilled water in the proportions recommended by the manufacturer. (See 5.1.)

4.3.5.1.2 *Stock cloth.* — Sheeting conforming to class A of Federal Specification CCC–S–271 shall be laundered and shall then be ironed on a flat-bed press as specified in method 5550 of Federal Specification CCC–T–191.

4.3.5.2 *Color.*—Five ml. of stock solution shall be added to 1,000 ml. of distilled water. The solution shall be mixed well and a portion of this solution shall be poured into a Nessler tube until it reaches the 100 ml. mark. The color of the solution shall be observed against a white background in good light.

4.3.5.3 *pH.*—The pH of the sample, as received, shall be determined at 25° ± 2° C., using a pH meter having a sensitivity and readability of 0.05 pH unit. A sealed type glass electrode shall be used in conjunction with a potassium chloride-calomel reference electrode. The pH meter shall be calibrated before use against a standard buffer solution of pH7.

4.3.5.4 *Insoluble dye.*—The sample, as received, shall be thoroughly mixed by shaking and 10 ml. of the solution shall be added by means of pipette, into a 1,500-ml. beaker, shall be diluted with 1,000 ml. of distilled water, and shall be stirred well. The solution shall be filtered through a Whatman No. 14 folder filter paper (32.0 cm.). The filter paper shall be visually observed for deposits of insoluble matter. (Readily and completely soluble dyes color the paper uniformly.)

4.3.5.5 *Freedom from pigment colors and methylene blue.*—Thirty-five ml. of distilled water shall be added to each of four 75-ml. test tubes containing 5 ml. of stock solution. One of these test tubes shall be retained as a control. To the others shall be added 10 ml. of 1:1 hydrochloric acid, 10 ml. of 10 percent sodium hydroxide, and 10 ml. of concentrated ammonium hydroxide, respectively. The solution shall be mixed well and the color changes shall be compared with table I.

4.3.5.6 *Resistance to decomposition by acid and alkali.*—Two ml. of stock solution shall be added, by means of pipette, to each of three 250-ml. beakers containing 100 ml. of distilled water buffered at pH 4.0, 7.0, and

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TABLE I.—*Color comparison*

Reagent	Aniline dye	Ultramarine blue	Prussian blue	Methylene blue
1:1 HCl	Turns slightly darker in color or turns emerald green	Decolorized	No action	Turns amber
10 percent NaOH	Turns purple or no action	No action	Decolorized	Turns reddish brown
NH_4OH (27 percent NH_3)	Turns lighter	No action	Partially decolorized	Turns reddish brown

8.0, respectively. The solution shall be stirred well and shall be allowed to stand for 24 hours. The solution shall be observed for evidence of decomposition such as precipitate or sediment.

4.3.5.7 *Fastness to ironing.*—A specimen of stock cloth approximately 8 by 8 inches in dimensions shall be prepared and shall be soaked for 5 minutes, at 90° ± 5° F., in 1,000 ml. of distilled water to which has been added 15 ml. of stock solution. The cloth shall be removed and shall be passed once through the wringer of a washing machine (do not wring by hand). The cloth shall be dried in air. The specimen shall be cut in half and one-half shall be ironed on a flat-bed press as specified in method 5550 of Federal Specification CCC–T–191. Using the other half as a control, color comparisons shall be made by visual observation to determine whether the sample has faded appreciably or turned yellow.

4.3.5.8 *Ease of removal in subsequent washing.* — Two specimens of stock cloth about 2½ by 4 inches in dimensions shall be cut. One specimen shall be soaked for 5 minutes, at 90° ± 5° F. in 1,000 ml. of distilled water to which has been added 15 ml. of stock solution. The cloth shall be dried in air. The other specimen of cloth shall be used as a control. Both specimens shall be washed separately, as specified in method 5610 of Federal Specification CCC–T–191. The bluing shall be removed completely from the sample cloth as determined by comparison with the control specimen.

4.3.5.9 *Tinctorial strength.*

4.3.5.9.1 *Apparatus (either one of the following may be used).*

4.3.5.9.1.1 *Colemann Universal Spectrophotometer, Model 11.*—The spectrophotometer shall be used with a 550 millimicron wave length setting, a PC-4 glass filter, and a 13 by 13 by 105 millimeter (mm.) square cuvette. It shall be adjusted to read 100 on the galvanometer using distilled water.

4.3.5.9.1.2 *Klett-Summerson Photoelectric Colorimeter.*—The colorimeter shall be used with a K.S. No. 54 glass filter (having a mean transmission of 540 millimicrons) and a 12.5 mm. test tube cell. It shall be adjusted to read zero, using distilled water as a blank.

4.3.5.9.2 *Procedure.* — Three hundred ml. of buffer solution adjusted to a pH of 4.0 shall be added to a 400-ml. beaker. By means of a pipette or burette, exactly 5 ml. of stock shall be added to the beaker, shall be stirred well, and shall be allowed to stand for exactly 5 minutes. At the end of 5 minutes, the color intensity of the solution shall be measured with either one of the previously standardized Coleman or Klett-Summerson instruments. The measurement shall be repeated using buffer solutions of pH 7.0 and 8.0.

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5. PREPARATION FOR DELIVERY

5.1 Labeling instructions for use.

5.1.1 *Type III, liquid bluing.*—Each bottle of liquid laundry bluing shall be marked durably and legibly as follows:

LIQUID LAUNDRY BLUING
(NONSOUR, SOUR)

Directions.—Prepare a stock solution of the bluing by diluting[1] fluid ounces of the concentrated blue with 5 gallons of fresh water. Rinse the container used to measure the blue with some of the water and add the rinsings to the stock solution. Store the stock solution in glass, enameled, stainless steel, or glazed earthenware containers; do not store in iron or galvanized metal.

Blue is added to the wash wheel as follows: (1) Use 2 ounces of blue stock solution per 50 pounds of load; (2) put the proper amount of blue stock solution in a pail and add water to dilute it still further, (3) add this pail full of blue to the load along the length of the washer while the cylinder, as seen through the supply door, is going *down.* If the bluing results are too light or too dark, the operator may vary the amount of stock solution used to obtain the desired tint.

5.2 Preservation and packaging.

5.2.1 *Packaging for domestic shipment—immediate use.*—Packaging shall be in accordance with manufacturer's commercial practice.

[1] Manufacturer shall insert here the number of fluid ounces (to the nearest ½ ounce) of his product which is to be used in preparing 5 gallons of stock solution.

Note: This bluing is concentrated and must be diluted before use as specified above. The bluing may be used in a sour bath or under nonsour conditions at the option of the operator. Generally, bluing in a sour bath gives better results.

5.3 Packing for domestic shipment.

—The subject commodity shall be packed in accordance with the Consolidated Freight Classification Rules, Postal Regulations, or Motor Freight Classification Rules, whichever may be applicable.

5.4 Marking.

5.4.1 *Interior containers.* — Unless otherwise specified in the contract or order, each container or package shall be plainly marked with the name of the article, the quantity contained herein, as defined by the contract or order under which shipment is made, and the name of the manufacturer.

5.4.2 *Exterior containers.* — Unless otherwise specified in the contract or order, shipping containers shall be marked with the name of the article, the quantity contained therein, as defined by the contract or order under which shipment is made, the name of the contractor and the number of the contract or order.

6. NOTES

6.1 Intended use.

6.1.1 *Type I, class A.*—Type I, class A bluing is intended for household or small batch bluing.

6.1.2 *Type I, class B.* — Type I, class B bluing is intended for use in laundries for flat work, and is not altered by the presence of dilute alkali.

6.1.3 *Type II.*—Type II bluing is intended for use in laundries for shirts, collars, and other starched work, and is not altered by the presence of dilute acids. It may also be used for flat work.

6.1.4 *Type III.* — Type III bluing is intended for general laundry use to neutralize the natural slightly yellow shade of fabrics by coloring them with a faint blue tint. The

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bluing works satisfactorily under acid, alkaline, or neutral conditions and may therefore be used whether or not souring is done.

6.2 Ordering data. — Purchasers should specify the following:

(a) Title, number, and date of this specification.
(b) Type and class required. (See 1.2.)
(c) Quantity per unit container. (See sec. 5 and 7.)
(d) Whether the subject commodity is to be prepared for domestic or overseas shipment. (See sec. 5 and 7.2.1.)
(e) Type and size of container required. (See 7.2.1.1.)
(f) Time of computing net weight. (See 6.3.)

6.3 Basis of purchase.

6.3.1 *Types I and II.*—Types I and II bluing should be purchased by net weight. Laundry bluing is subject to a possible gain or loss of weight, depending upon packaging, atmospheric, or storage conditions. Therefore, the time of computing net weight with reference to acceptance or delivery should be specified in the contract or order.

6.4 Basis for award, type III.—The basis for award should be determined by the following formula:

$$\text{(Delivered price per quart)} \times \frac{A}{5}$$

where A = number of fluid ounces of bluing recommended to prepare 5 gallons of stock solution. (See 5.1.)

6.5 Transportation description.

6.5.1 *The applicable transportation description and carload and truckload minimum weights to be added are as follows:*

Types I and II.
Bluing, laundry, dry.
Carload weight 30,000 pounds.
Bluing, laundry, dry, not otherwise indexed.
Truckload weight 30,000 pounds.

Type III.
Bluing, laundry, liquid.
Carload weight 36,000 pounds.
Bluing, laundry, not otherwise indexed, liquid.
Truckload weight 30,000 pounds.

Patent notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

7. DEPARTMENTAL REQUIREMENTS

7.1 The following specifications and standard, of the issues in effect on date of invitation for bids, and special requirements, form a part of this specification for purchases made under this specification by the respective departments:

7.2 Army, Navy, and Air Force.

Federal Specifications:
U-M-186—Medicinal Products and Clinical Laboratory Reagents; General Specifications for Containers (Packaging and Packing).
NN-B-621—Boxes, Wood, Nailed and Lock-Corner.
LLL-B-631—Boxes, Fiber, Corrugated (for Domestic Shipment).
LLL-B-636—Boxes, Fiber, Solid (for Domestic Shipment).

Military Specifications:
JAN-P-105—Packaging and Packing for Overseas Shipment—Boxes, Wood, Cleated, Plywood.

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JAN–P–106—Packaging and Packing for Overseas Shipment—Boxes, Wood, Nailed.

JAN–P–108—Packaging and Packing for Overseas Shipment—Boxes, Fiberboard (V-Board and W-Board), Exterior and Interior.

JAN–P–120—Packaging and Packing for Overseas Shipment — Cartons, Folding, Paperboard.

MIL–C–124—Containers (Cans, Pails and Drums), Metal (for Other Than Subsistance Items).

MIL–P–10225—Packaging of Glass Containers, One-Gallon Capacity and Smaller, for Other Than Medicinal Products for Domestic Shipment.

MIL–L–10547—Liners, Case, Waterproof.

Military Standard:

MIL–STD–129—Marking of Shipments.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

7.2.1 Preparation for delivery.

7.2.1.1 *Packaging for domestic shipment and storage or overseas shipment.*

7.2.1.1.1 *Types I and II.*—The bluing shall be packaged in 1-pound or 5-pound cartons or cans, as specified. (See 6.2). The cartons shall be sift-proof, conforming to style 1, type A, class a or b of Military Specification JAN–P–120. The cans shall conform to type IV of Military Specification MIL–C–124. If the product is for overseas shipment, covers of cans shall be spot soldered to the friction ring at three points equidistant from each other around the periphery of the cover.

7.2.1.1.2 *Type III.*—Liquid bluing shall be furnished in 1-quart or 1-gallon round flint glass bottles as specified (see 6.2) conforming to type I, class B of Federal Specification U–M–186 for both the bottles and closures. Closures shall be plastic, continuous thread, screw type. A secondary closure shall be applied over the screw cap and extended down on the neck of the bottle. Secondary closure shall be of a self-shrinking cellulose band seal to provide a leak-proof closure and to prevent any possibility of opening during shipment. Alternatively, secondary closure may be made by the application of stretchable, pressure-sensitive tape of minimum ¾ inch width. When applied, tape shall overlap itself a minimum of 1 inch.

7.2.1.2 *Packing.*

7.2.1.2.1 *Types I and II.*

7.2.1.2.1.1 *For domestic shipment and storage.*—Forty-eight 1-pound packages or cans or twelve 5-pound packages or cans shall be packed in regular-slotted-style fiberboard boxes conforming to Federal Specifications LLL–B–631 or LLL–B–636, or nailed wood boxes conforming to style 4 of Federal Specification NN–B–621. The bursting strength of the fiberboard shall be not less than 275 pounds. Closure of containers shall be in accordance with the applicable container specification.

7.2.1.2.1.2 *For overseas shipment.*—Forty-eight 1-pound packages or cans or twelve 5-pound packages or cans shall be packed in fiberboard boxes or nailed wood boxes conforming to style RSC, symbol V3s or V3c of Military Specifications JAN–P–108 or JAN–P–106.

7.2.1.2.1.3 *Liner.*—Each shipping container shall be provided with a case liner conforming to Military Specification MIL–L–10547 and shall be sealed in accordance with the appendix thereto.

7.2.1.2.2 *Type III.*

7.2.1.2.2.1 *For domestic shipment and storage.*—Bottles of liquid bluing packaged

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as specified in 7.2.1.1.2 shall be packed in a corrugated fiberboard box provided with slotted partitions, pads, and liners in accordance with Military Specification MIL–P–10225.

7.2.1.2.2.2 *For overseas shipment.*—Bottles of bluing, packed as specified in 7.2.1.2.2.1, shall be overpacked in cleated plywood or nailed wood boxes conforming to Military Specifications JAN–P–105 and JAN–P–106, respectively. Fiberboard boxes shall be lined with a sealed caseliner conforming to Military Specification MIL–L–10547 and closed and strapped in accordance with the appendix to the applicable box specification.

7.2.1.3 *Marking.*—In addition to any special marking required by the contract or order, unit and intermediate packages and exterior shipping containers shall be marked in accordance with Military Standard MIL–STD–129. The nomenclature shall be as follows:

Bluing, Laundry.

MILITARY INTEREST:
Army—Q M
Navy—Sh *MC*
Air Force.

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